

1. Administrative & Study Identification

Full Study Title: A Phase 1B/2, Open Label Umbrella Study of Elranatamab (PF-06863135), A B-Cell Maturation Antigen (BCMA) CD3 Bispecific Antibody, in Combination with Other Anti-Cancer Treatments In Participants With Multiple Myeloma **Short Title/ Acronym:** MagnetisMM-4: Umbrella Study of Elranatamab (PF-06863135) in Combination With Anti-Cancer Treatments in Multiple Myeloma **Protocol Number:** C1071004 **NCT Number:** NCT05090566 **Posting ID:** 518791751437126444 **Sponsor Name:** Pfizer **Investigational Product:** Elranatamab (PF-06863135) in combination with other anti-cancer treatments (e.g., niraparic acid, lenalidomide, dexamethasone) **Phase of Development:** Phase 2 (Phase 1B/2) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety and tolerability of elranatamab in combination with other anti-cancer therapies in participants with Relapsed/Refractory Multiple Myeloma (RRMM) to determine the Recommended Phase 2 Dose (RP2D) and clinical benefit. **Secondary Objectives:** To evaluate overall response rate (ORR), complete response (CR) rate, time to response, duration of response, progression-free survival (PFS), overall survival (OS), minimal residual disease (MRD) negativity rate, pharmacokinetics (PK), and immunogenicity.

2.2 Background & Rationale To address the clinical need in patients with relapsed or refractory multiple myeloma who have progressed on standard therapies. Elranatamab is a humanized bispecific molecule targeting BCMA on myeloma cells and CD3 on T-cells. This umbrella study evaluates combinations of elranatamab with agents like gamma-secretase inhibitors (e.g., niraparic acid) or immunomodulatory drugs (IMiDs) to potentially enhance BCMA-directed therapy and overcome resistance mechanisms.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Study) **Control Type:** Single-arm (Dose escalation and expansion cohorts per sub-study) **Blinding:** Open-label **Randomization:** Non-randomized

3.2 Planned Enrollment & Duration Target N: 34 evaluable patients **Number of Sites:** Multicenter **Estimated Study Start:** October 2021 **Current Trial Status:** Active, not recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with Relapsed/Refractory Multiple Myeloma (RRMM). 2. At least 3 prior lines of therapy; refractory to at least one IMiD, one proteasome inhibitor, and one anti-CD38 antibody. 3. Measurable disease defined by at least one of the following: Serum M-protein ≥ 0.5 g/dL by SPEP, Urinary M-protein excretion ≥ 200 mg/24 hours by UPEP, or Serum immunoglobulin FLC ≥ 10 mg/dL (≥ 100 mg/L) AND abnormal serum immunoglobulin kappa to lambda FLC ratio. 4. ECOG performance status 0 - 1. 5. Resolved acute effects of any prior therapy to baseline severity or CTCAE Grade ≤ 1 .

4.2 Exclusion Criteria 1. Active plasma cell leukemia, amyloidosis, or POEMS syndrome. 2. Stem cell transplant within 12 weeks prior to enrollment, or active graft-versus-host disease (GVHD). 3. Any active uncontrolled bacterial, fungal, or viral infection, or impaired cardiovascular function or clinically significant cardiovascular diseases within 6 months prior to enrollment. 4. Previous administration with an investigational drug within 30 days or 5 half-lives preceding the first dose of study treatment (whichever is longer). 5. Previous treatment with BCMA bispecific antibody (Sub-Study A Only). 6. Previous treatment with BCMA directed therapy (Sub-Study B Only).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Elranatamab administered with a 2-step-up priming dose schedule to mitigate the risk of cytokine release syndrome (CRS), followed by therapeutic dose administration. Administered in separate sub-studies (e.g., Sub-study A: combined with nirogacestat; Sub-study B: combined with lenalidomide and dexamethasone). **Route of Administration:** Subcutaneous (SC) for elranatamab; combined with specified oral or intravenous partner therapies. **Duration of Treatment:** Cycles continue until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for multiple myeloma, including necessary infection prophylaxis and hematologic support. **Prohibited:** Previous administration of an investigational drug within 30 days or 5 half-lives preceding the first dose. Live attenuated vaccines are generally restricted.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Disease Assessment (SPEP/UPEP/sFLC), Safety Labs

| | **Baseline** | Day 0 | Enrollment, First Priming Dose, Pharmacokinetics (PK), Vitals | | **Interim** | Treatment Cycles | Safety Labs, CRS Monitoring, Efficacy Assessment, Dose Escalation/Expansion evaluation | | **End of Study** | Disease Progression | Final Vital Signs, Exit Interview, Follow-up for Survival / Subsequent Therapy |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Sub-Study A Phase 1 (Dose Limiting Toxicity)**: Number of participants with Dose Limiting Toxicity. 2. **Sub-Study A Phase 2 (Objective Response Rate)**: Objective response rate (IMWG response criteria). 3. **Sub-Study B Phase 1 Escalation (Dose Limiting Toxicity)**: Number of participants with Dose Limiting Toxicity.

7.2 Secondary & Exploratory Endpoints 1. **Sub-Study A Phase 1**: Objective Response Rate. 2. **Sub-Study A Phase 1 and Phase 2**: Complete Response Rate. 3. **Sub-Study A Phase 1 and Phase 2**: Time to Response. - Additional exploratory endpoints include duration of response, Progression-Free Survival (PFS), Overall Survival (OS), Minimal residual disease (MRD) negativity rate, and immunogenicity/PK assessments.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard intensive clinical and laboratory monitoring with specific focus on Cytokine Release Syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS). **Serious Adverse Events (SAEs)**: Reported within 24 hours of site awareness to the sponsor. **Data Monitoring Committee (DMC)**: Independent Data Monitoring Committee (IDMC) utilized to oversee safety and trial progression.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive ≥ 1 dose of study drug); Efficacy Set (participants meeting inclusion criteria and receiving treatment). **Sample Size** **Justification**: Targeted enrollment of 34 patients; a Bayesian logistic regression model is utilized during Phase 1 for continual dose escalation reassessment. **Handling of Missing Data**: Efficacy endpoint calculations and missing PK data will be handled via standard imputation parameters per the finalized SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Sponsor-led onsite and remote clinical monitoring visits to verify source data and safety events.

Audit Trail: eCRF entry logs, data clarification tracking, and secure electronic data capture (EDC) systems.

11. Study Identifiers & Governance

Primary ID: C1071004 **Posting ID:** 518791751437126444 **Registry Number:** NCT05090566 **Sponsor/Lead Organization:** Pfizer **Company:** Pfizer **Study Title:** MagnetisMM-4: Umbrella Study of Elranatamab (PF-06863135) in Combination With Anti-Cancer Treatments in Multiple Myeloma **Official Regulatory Title:** A PHASE 1B/2, OPEN LABEL UMBRELLA STUDY OF ELRANATAMAB (PF-06863135), A B-CELL MATURATION ANTIGEN (BCMA) CD3 BISPECIFIC ANTIBODY, IN COMBINATION WITH OTHER ANTI-CANCER TREATMENTS IN PARTICIPANTS WITH MULTIPLE MYELOMA

12. Clinical Framework (Multiple Myeloma Specific)

Primary Condition: Multiple Myeloma (Relapsed/Refractory Multiple Myeloma [RRMM]) **Diagnosis Threshold:** Measurable disease defined via SPEP ≥ 0.5 g/dL, UPEP ≥ 200 mg/24h, or sFLC ≥ 10 mg/dL with abnormal ratio **Medical Methodology:** Bispecific Antibody targeting BCMA and CD3 + Standard Anti-Cancer combinations (including dexamethasone) **Optimization Target:** Selection of Recommended Phase 2 Dose (RP2D) and optimization of Objective Response Rate (ORR)

13. Study Procedures & Methodology

Management Pathway: Gradual 2 step-up priming dose strategy for optimal Cytokine Release Syndrome (CRS) mitigation. **Education Protocol (HCP):** Comprehensive protocol training with a focus on CRS and neurotoxicity identification/management. **Education Protocol (Patient):** Strict onsite monitoring instructions during priming steps; continuous self-monitoring for infection and CRS symptoms. **Quality Audit Mechanism:** Routine sponsor oversight, safety data reviews by the IDMC, and protocol adherence evaluations.

14. Observation & Follow-Up Schedule

Total Duration: Continuous until disease progression or unacceptable toxicity. **Onsite Visit Cadence:** High frequency

during Cycle 1 (priming and safety monitoring); regular interval follow-ups per subsequent treatment cycles. **Remote Monitoring Frequency:** Regular interval survival and progression follow-up post-treatment discontinuation. **Checklist Review Interval:** Baseline and start of every cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				